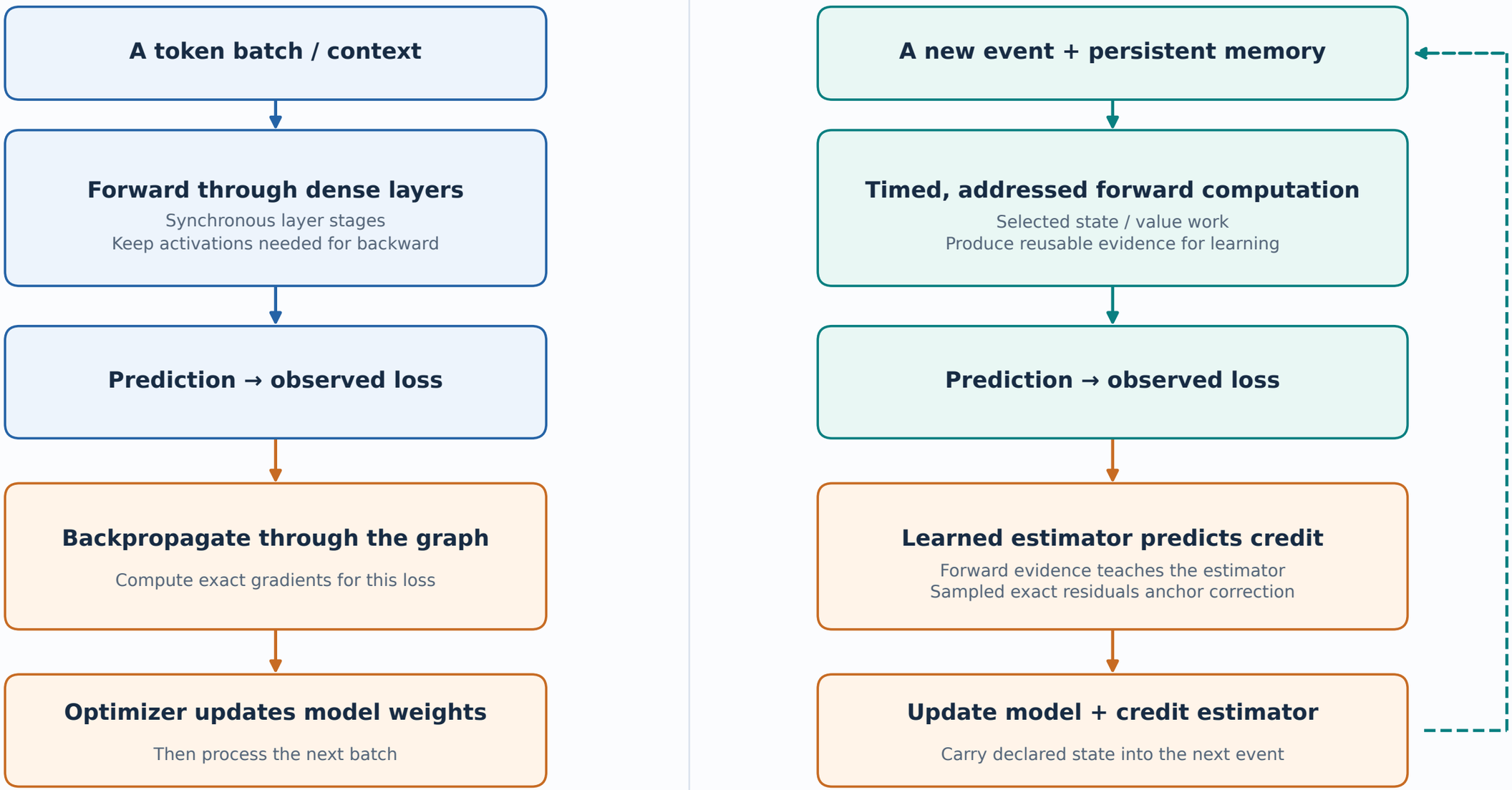
Same goal. A different way to assign learning credit.

Ordinary synchronous Transformer training vs the proposed temporal, sparse, learned-credit substrate



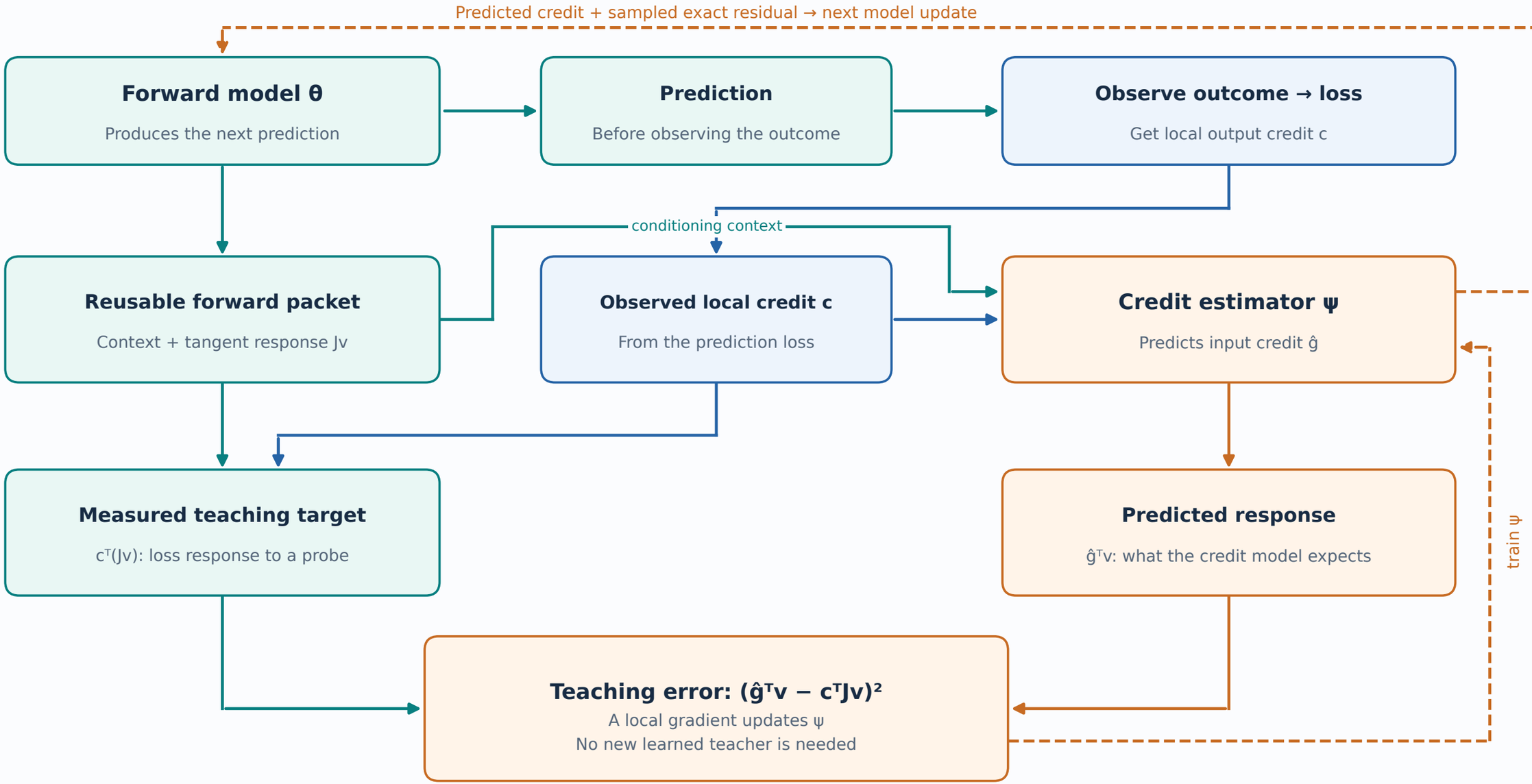
The intended saving: reuse learned credit instead of recomputing all learning work at every event.

Evidence scope: local/deep numerical contracts and measured component savings; the complete sparse, persistent loop is in development.

Charge discovery, counterfactuals, teaching, audits, state transport and optimizer work. No general Transformer advantage is claimed here.

How the forward pass teaches the credit estimator

One local credit boundary; measured responses and observed outcomes close the training loop



Why it closes: the target is a response measured from the forward computation, anchored on an observed loss.

Residual audit: $\hat{g} + I/\pi \cdot (g - \hat{g})$, with positive audit probability π, preserves the conditional raw-gradient mean.

For content + time, use joint probes. Shared producer probes retain their cross terms. Versions and causal horizons must match.

This panel explains continuous credit. Hard-route interventions use future-loss advantages; their acquisition and audits are additional work.